Free Pell coordinates and a smaller nine-variable equation for ZFC consistency

Exact degree 4 686 812; 560 binary arithmetic operations

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Abstract

We construct a fully specified integer-coefficient polynomial in nine strictly positive integer unknowns whose zero-existence condition is equivalent to the repository's fixed 31-variable refutation system. Composing with its stated ordinary compiler and K215 calibration theorems gives the desired ZFC-provable equivalence between absence of zeros and Con(ZFC). The new polynomial has exact total degree 4 686 812 and a division-free evaluator with 560 live binary operations. Against the lowest-degree nine-variable candidate found in the incoming archives, 7 344 328 in 738 operations, the reductions are 36.18% in degree and 24.12% in arithmetic size.

The principal new idea is to replace a degree-deg R Pell coordinate by a free positive coordinate. A positive-margin condition already implied by the final norm reconstructs its old lower bound. A small quotient translation makes that reasoning valid before the margin is known to be positive. A second change removes an unnecessary leading binary block from the carry code. A shared evaluator for twenty of the twenty-two quadratic rows supplies most of the operation saving. We also give fifteen-positive-variable equations with 73 operations at degree 54 and 74 operations at degree 46.

All new transformations are proved as ordinary mathematics. Exact coefficient data, literal circuits, source bindings, and reproducible finite checks accompany the paper. The result is not a claim of global minimality or a new end-to-end Lean/Rocq certificate; the inherited compiler dependency and the distinction from an object-language ZFC derivation are kept explicit.

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1 Result, comparison, and meaning of "simpler"

There is no known single ordering that captures simplicity of Diophantine equations. Variable count, total degree, arithmetic circuit size, coefficient size, and the length of a correctness proof can move in different directions. This note improves the degree and arithmetic size together at the nine-positive-variable point. It does not claim to reduce the established nine-natural-variable existence bound to eight, nor to give an optimal polynomial among all representations.

The comparison is frozen at ProveIt commit `4b4349064a750ba6347293b55874a7f92abdd0db`. Both the placed collection and all eight incoming ZIP archives were inspected. This matters: several incoming improvements are absent from the active short circuit documentation. In particular, the correct short-operation comparator is 66, not 67. Table 1 distinguishes those dimensions.

Construction	Witness domain	Exact degree	Operations
Placed package 45	$\mathbb{Z}^9_{>0}$	129 875 980	788
Incoming selective quadratic	$\mathbb{Z}^9_{>0}$	7 344 328	738
Incoming sparse Pell, fastest listed	$\mathbb{Z}^9_{>0}$	15 502 748	548
This note, primary	$\mathbb{Z}^{\mathbf{9}}_{\mathbf{>0}}$	**4 686 812**	**560**
This note, original-H alternative	$\mathbb{Z}^9_{>0}$	4 686 812	562
This note, nonnegative translate	$\mathbb{N}^9$	4 686 812	567
Incoming squared quotient	$\mathbb{Z}^{15}_{>0}$	157	66
Incoming squared quotient SOS	$\mathbb{Z}^{15}_{>0}$	80	72
This note, Q_{54}	$\mathbb{Z}^{\mathbf{15}}_{\mathbf{>0}}$	**54**	**73**
This note, Q_{46}	$\mathbb{Z}^{\mathbf{15}}_{\mathbf{>0}}$	**46**	**74**
Incoming gap minimum	$\mathbb{Z}^{15}_{>0}$	44	75

Table 1: Audited comparison scope. The nine-variable primary improves the lowest-degree incoming point in both columns. The 548-operation incoming point remains a different tradeoff. No unrestricted-integer assertion is made for the positive-domain polynomials.

Let $\mathsf{Ref}_{\mathrm{ZFC}}$ mean that there is a finite syntactic derivation of falsity from the repository's ZFC axiom predicate, and let Con(ZFC) be its negation. Our main, entirely local mathematical result is a constructive equivalence between the new polynomial and the specified finite source system. The interpretation of that source system as $\mathsf{Ref}_{\mathrm{ZFC}}$ is the named inherited compiler theorem, not an inference from sampled evaluations.

Theorem 1.1 (Construction and endpoint)**.** *Let $p_1, \dots, p_{22}$ be the fixed quadratic polynomials of Appendix B, with the complete package-31 coefficient tuple specified in Section 3. The polynomial $\mathcal{P}$ defined below belongs to $\mathbb{Z}[b, g, h, s, w, C, i, j, n]$ and satisfies*

$$\exists\,(b,g,h,s,w,C,i,j,n) \in \mathbb{Z}^9_{>0} :\ \mathcal{P} = 0 \quad\Longleftrightarrow\quad \exists\,\mathbf{x} \in \mathbb{N}^{31} :\ p_1(\mathbf{x}) = \cdots = p_{22}(\mathbf{x}) = 0.$$

It has exact total degree 4 686 812 and the supplied primary circuit has 332 multiplications, 173 additions, and 55 subtractions. Using the repository's ordinary source-compiler and K215 calibration theorems, the equivalence is formalizable in ZFC and yields

$$\boxed{\mathrm{ZFC} \ \vdash\ \left[\neg\exists\,\mathbf{u} \in \mathbb{Z}^9_{>0} : \mathcal{P}(\mathbf{u}) = 0\right] \ \longleftrightarrow\ \mathrm{Con(ZFC)}.}$$

The nine variables in this statement are the only quantified unknowns. Every other symbol introduced below is a polynomial abbreviation or a fixed integer. Fixed powers are expanded by multiplication in the literal circuit. The theorem does not assume consistency and does not decide whether the equation has a zero.

2 The explicit equation

2.1 Fixed source data and carry polynomials

Put

$$\kappa = 2\,134\,940, \qquad \lambda = 2^{\kappa}, \qquad \chi = 2^{16\,012\,039}, \qquad B = \lambda b^2.$$

The fixed integer-coefficient polynomial D_{c} and the two fixed lists $w_1, \dots, w_{31}$ and $t_1, \dots, t_{22}$ are defined in Section 3. Set

$$\ell = \sum_{r=1}^{31} B^{w_r}, \qquad A_{\mathrm{c}} = \sum_{r=1}^{22} B^{t_r}, \qquad N_0 = B^{184}, \tag{1}$$

$$M_{\mathrm{c}} = N_0 - 1 - (b-1)\ell, \qquad J_{\mathrm{c}} = 2(1+g)^2 D_{\mathrm{c}}(B) + A_{\mathrm{c}}, \tag{2}$$

$$S_{\mathrm{c}} = g + N_0 J_{\mathrm{c}}, \qquad T_{\mathrm{c}} = M_{\mathrm{c}} + N_0(B-2)A_{\mathrm{c}}, \tag{3}$$

$$L = bB^{5741}, \qquad R = (S_{\mathrm{c}} + T_{\mathrm{c}})L + L - 1 - S_{\mathrm{c}}, \tag{4}$$

$$N = \chi B^{2966}, \qquad G_{\mathrm{cap}} = 32bB^{183}. \tag{5}$$

Here L is the binary block length and N is the threshold supplied to Jones's ratio theorem. They are different polynomials. In particular, the old leading L^2 term is absent from R.

2.2 Pell registers and the free coordinate

The coordinate C in the following display is a free positive unknown. It is not defined to be $2R + 1 + bw + \phi$.

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M_0 &= RY,\\
P &= 2M_0^2 U, & K &= R + 1 + h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R+1, & W &= bw, & J &= 4A - 5,\\
Q &= 3WC - 2(W^2-1), & D &= (A^2-1)C^2 + 1,\\
E &= iJDC^2, & F &= (A^2-1)E^2 + 1,\\
G &= A + F(CD + 1 - A), & H &= C + F\big(B_1 + (j-1)C\big),\\
I &= (G^2-1)H^2 + 1.
\end{aligned} \tag{6}$$

The use of $j - 1$ is intentional: the CRT-absorbed parameter ranges over $\mathbb{N}$, while the displayed nine-variable presentation uses only positive inputs. This shift is charged in the 560-operation ledger.

Define

$$\begin{aligned}
a_0 &= (P^2-1)K^2 + 1, & a_1 &= 64DFI,\\
m &= \big[K^2 - 4(C - KY)^2\big]G_{\mathrm{cap}} - 8gK^2,\\
k_* &= iJC, & T_* &= k_*(8D^2 - 4D - 1),\\
V_0 &= PK, & V_1 &= T_* GH,\\
\Omega &= Jm, & \Theta &= J(n + W^2) + Q - \Omega(V_0 + V_1).
\end{aligned} \tag{7}$$

The requested equation is

$$\boxed{\mathcal{P} = \big[\Theta^2 - \Omega^2(a_0 + a_1)\big]^2 - 4\Omega^4 a_0 a_1 = 0.} \tag{8}$$

Equations (1)–(7) are definitions, not additional equations with independent unknowns. Thus (8) is one ordinary polynomial equation in nine unknowns. The fixed coefficient recipe contains no consistency-dependent selection, unbounded proof search, variable exponent, or unspecified r.e. index.

2.3 A slightly longer form with a simpler congruence interface

For the 562-operation alternative, replace H in (6) by $H = B_1 + jC$ and put

$$u = JF, \quad v = QF + J(H - C), \quad \Omega = um, \quad \Theta = u(n + W^2) + v - \Omega(V_0 + V_1).$$

Keep the same norm (8). Every input then appears directly as a positive coordinate. The degree is unchanged. This version is useful for checking the precise connection to the original Jones ratio theorem; the two presentations are existence-equivalent, not claimed to be the same polynomial on the same tuples.

3 Coefficient binding and finite selective code

3.1 A coherent, materialized tuple

The four source ports are β, K_0, M_C, M_F. We use the entire `wang943-mixed-121x8801` tuple from placed package 31, including the already shifted M_F port. The archive contains all four compressed little-endian magnitude payloads, their hashes, the original binding manifest, and the coefficient-construction proof. Their bit lengths are

$$1\,064\,921, \quad 1\,070\,004, \quad 1\,064\,921, \quad 1\,064\,922.$$

In particular $\beta = 2^{1\,064\,921} - 1$. The other three fixed integers are given by those finite payloads and their reproducible native compiler recipe. An expanded decimal printing is unnecessary for a complete definition and would obscure the construction. The supplied data do not leave any port to be chosen later.

Let $p_r(\mathbf{x})$ be the residual in row r of Appendix B after every one of its positive source coordinates is replaced by $x_i + 1$. For $|\alpha| \le 2$, define

$$c_{r,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i! \, [\mathbf{x}^\alpha] p_r(\mathbf{x}).$$

There are 136 nonzero row-monomial records, using 71 distinct monomials. Each $c_{r,\alpha}$ is an explicitly supplied polynomial in the four fixed ports. Bounding a recipe $\sum_e a_e \prod_s q_s^{e_s}$ by $\sum_e |a_e| 2^{\sum_s e_s \, \mathrm{bitlength}(q_s)}$ gives

$$|c_{r,\alpha}| < M_{\mathrm{coef}} = 2^{2\,134\,927}, \qquad \lambda = 8192 M_{\mathrm{coef}}.$$

The finite verifier reconstructs the shifted rows and these recipes from the literal source, rather than trusting the declared counts.

3.2 The selected weights and row positions

In the variable order of Appendix B, use

$$(w_1, \ldots, w_{31}) = (9, 25, 23, 139, 35, 151, 168, 21, 91, 181, 94, 52, 150, 180, 7, 135, \\ 38, 88, 183, 106, 178, 136, 20, 138, 8, 64, 105, 154, 37, 137, 120).$$

Their maximum is 183. Assign targets to rows as follows:

Row	16	2	9	14	21	7	15	22	13	3
Target	70	86	451	685	1050	1296	1530	1874	2212	2358
Row	19	20	1	10	17	8	4	6	11	18
Target	2696	2780	2792	3157	3290	3634	3929	4238	4384	4527
Row	12	5								
Target	4865	5190								

Write t_r for the target assigned to row r, and $e(\alpha) = \sum_i \alpha_i w_i$. Define the fixed polynomial

$$D_{\mathrm{c}}(z) = \sum_{r,\alpha} c_{r,\alpha} z^{t_r - e(\alpha)}. \tag{9}$$

Every exponent is nonnegative. The leading exponent is 5190 and its coefficient is exactly 2.

The certificate checks, for every supported pair (r, α) and every multi-index γ of degree at most two,

$$t_r - e(\alpha) + e(\gamma) = t_s \quad \Longleftrightarrow \quad r = s \text{ and } \alpha = \gamma. \tag{10}$$

There are $136 \cdot 528 = 71\,808$ interactions. This is a complete finite check of a stated combinatorial condition, not a probabilistic test. It implies the coefficient identity

$$[z^{t_r}] \left(1 + \sum_i x_i z^{w_i}\right)^2 D_{\mathrm{c}}(z) = 2p_r(\mathbf{x}). \tag{11}$$

Indeed the coefficient of a padded monomial in the square is $2!/((2-|\alpha|)!\prod_i \alpha_i!)$, which cancels the factorial weight. The selective code is inherited from the incoming selective-quadratic archive; its reuse is not claimed as a new combinatorial discovery.

4 A smaller carry code: remove the leading block

4.1 Bounds valid before any equation holds

For all positive b, $B \geq \lambda$, so $B - 1 > M_{\mathrm{coef}}$ and $B > 128$. The positive leading coefficient of D_{c} dominates all its lower signed coefficients:

$$B^{5190} < D_{\mathrm{c}}(B) < \frac{M_{\mathrm{coef}} B^{5191}}{B-1}.$$

The digits of M_{c} are $B - b$ at the 31 selected weights and $B - 1$ elsewhere in its 184-digit range. Thus $0 \leq M_{\mathrm{c}} < N_0$, $J_{\mathrm{c}}, S_{\mathrm{c}} > 0$, and $T_{\mathrm{c}} \geq 0$. In particular

$$R = (L-1)S_{\mathrm{c}} + L(T_{\mathrm{c}} + 1) - 1 \geq L - 1 \geq \max(8, b). \tag{12}$$

Also $N \geq \max(8, b)$ and $G_{\mathrm{cap}} > 0$. These unconditional facts are what the later ratio argument needs. No inequality $R \geq N$ is assumed.

On the branch $g < 4bB^{183}$, integrality gives $1 + g \leq 4bB^{183}$, and $g < N_0$ because $B > 4b$. Put $Z = B^{5557}$. Then

$$2(1+g)^2 D_{\mathrm{c}}(B) < \frac{32 M_{\mathrm{coef}} b^2 B^{5557}}{B-1} < \frac{Z}{128}, \qquad A_{\mathrm{c}} < \frac{Z}{128},$$

so $0 < J_{\mathrm{c}} < Z/64$. Consequently

$$S_{\mathrm{c}} < N_0(Z/64 + 1) < B^{5741}/32.$$

Since $(B-2)A_{\mathrm{c}} < B^{5191}$ and $B^{366} > 64$,

$$T_{\mathrm{c}} < N_0(1 + B^{5191}) < B^{5741}/32.$$

It follows that

$$0 \le S_{\mathrm{c}}, T_{\mathrm{c}}, \qquad S_{\mathrm{c}} + T_{\mathrm{c}} < L = bB^{5741}, \tag{13}$$

including the boundary $b = 1$.

4.2 The carry test detects precisely the source zeros

Suppose b is a power of two; then so is B. The low binary mask says

$$g \,\&\, M_{\mathrm{c}} = 0 \quad \Longleftrightarrow \quad g = \sum_i x_i B^{w_i}, \qquad 0 \le x_i < b.$$

The coefficient at the padding position zero is fixed at one by $(1+g)^2$; zero is not one of the variable weights.

For such g, put $h_t = [z^t](1 + \sum_i x_i z^{w_i})^2 D_{\mathrm{c}}(z)$. The coefficient bound gives

$$|h_t| < M_{\mathrm{coef}}(1 + \textstyle\sum_i x_i)^2 < 1024 M_{\mathrm{coef}} b^2 = B/8.$$

Before ordinary radix normalization, the digits of J_{c} are $2h_t + \delta_t$, where $\delta_t = 1$ at a target and zero elsewhere. An incoming borrow is always 0 or -1. At a target, a zero h_t therefore leaves the normalized digit 0 or 1. A positive h_t leaves a digit at least 2 and below $B/2$; a negative h_t leaves a digit above $B/2$. The mask $B-2$ accepts exactly the first case. This proves

$$J_{\mathrm{c}} \,\&\, (B-2)A_{\mathrm{c}} = 0 \quad \Longleftrightarrow \quad h_{t_r} = 0 \text{ for all } r.$$

The borrow assertion follows directly by induction from $|h_t| < B/8$; it does not use a bit operation on negative integers. The evaluated J_{c} is already positive and below Z.

The low and high blocks in (3) do not overlap. Using (11), we obtain

$$S_{\mathrm{c}} \,\&\, T_{\mathrm{c}} = 0 \quad \Longleftrightarrow \quad g \text{ encodes a common nonnegative zero of all } p_r. \tag{14}$$

Conversely, for a source zero choose any sufficiently large power of two b and use its radix encoding g. The source has $p_5(\mathbf{0}) = 1$, so that encoding is positive. Moreover $g < bB^{183}$ and therefore $32g < G_{\mathrm{cap}}$, the stronger bound needed for the completeness direction of Jones's theorem.

4.3 Exact two-block valuation and its parity

Lemma 4.1 (Two blocks without a leading sentinel)**.** *Let $L = 2^u$ and $S, T \ge 0$ with $S + T < L$. Set*

$$R = (S+T)L + (L - 1 - S).$$

Then

$$v_2\binom{2R}{R} = u + \mathrm{pop}(T) - \big(\mathrm{pop}(S) + \mathrm{pop}(T) - \mathrm{pop}(S+T)\big).$$

In particular $2^{u+\mathrm{pop}(T)} \mid \binom{2R}{R}$ if and only if $S \& T = 0$.

Proof. The two binary blocks do not overlap, and the low block is the u-bit complement of S. Thus $\mathrm{pop}(R) = \mathrm{pop}(S+T) + u - \mathrm{pop}(S)$. Legendre's identity $v_2(r!) = r - \mathrm{pop}(r)$ gives $v_2\binom{2R}{R} = \mathrm{pop}(R)$. Finally $\mathrm{pop}(S) + \mathrm{pop}(T) - \mathrm{pop}(S+T)$ is nonnegative and vanishes exactly when the addition has no carries. This proves both assertions. □

Write $b = 2^k$ and $B = 2^v$, so $v = \kappa + 2k$. The low mask has $184v - 31k$ set bits, and the 22 high target digits each have $v - 1$ set bits. Therefore

$$\operatorname{pop}(T_{\mathrm{c}}) = 206v - 31k - 22, \qquad u = \log_2 L = k + 5741v.$$

Their sum is

$$u + \operatorname{pop}(T_{\mathrm{c}}) = 5947\kappa + 11864k - 22 = 2\log_2(\chi B^{2966}). \tag{15}$$

The exponent of χ is $15\kappa/2 - 11 = 16\,012\,039$. Together, Lemma 4.1 and (14) show that the source is represented by the three numerical requirements

$$b \text{ is a power of two}, \qquad N^2 \mid \binom{2R}{R}, \qquad g < 4bB^{183}.$$

The incoming comparator used $2bB^{5741}$ and an additional L^2 term. Deleting that term and halving the block size changes the threshold by two binary places, explaining the factor-two change in χ. The proof above checks this exact change; it is not a heuristic degree estimate. At $b = 1$ the low mask would force $g = 0$, so the enlarged positive parameter domain adds no zero.

5 The free-coordinate theorem

5.1 The numerical interface imported from Jones

We use the numerical ratio theorems, not a black-box application of the hard-wired universal-polynomial formula. In the repository their precise interfaces are `Jones1982.Nine.ratio_sound` and `ratio_complete` in `NineRatio.lean`, corresponding to the Pell and binomial analysis of Jones [1]. With $R, N \geq 8$, $1 \leq b \leq \min(R, N)$ and $G_{\mathrm{cap}} > 0$, the standard registers use

$$C = B_1 + W + \phi, \quad H_{\mathrm{old}} = B_1 + j_{\mathrm{old}}C, \qquad \phi \geq 0, \quad j_{\mathrm{old}} > 0.$$

Their four conditions are

$$a_0 = \square, \qquad DFI = \square, \qquad JF \mid QF + J(H_{\mathrm{old}} - C), \qquad m > 0. \tag{16}$$

The soundness theorem implies a power-of-two b, the central-binomial divisibility, and $8g < G_{\mathrm{cap}}$. The completeness theorem supplies witnesses from the same power and divisibility conditions when $32g < G_{\mathrm{cap}}$. These asymmetric bounds are precisely those established in the preceding section. All source data satisfy the unconditional size hypotheses.

The next proof explains why the lower-bound slack ϕ can be removed from the actual polynomial. Recovering it after using the ratio theorem would be circular. We instead recover it from the final norm before that theorem is invoked.

5.2 Unconditional positive majorants

Assume only the allowed coordinate domains and the unconditional source bounds. Then $U, Y \geq 64$, $K \geq P + R > 1$, and $A \geq 2$. Moreover

$$A = RY(U + 1) > U = N^2 w \geq bw = W.$$

Because $C \geq 1$ is a positive independent input,

$$D \geq A^2 > W^2, \qquad D > C^2, \qquad WC < D.$$

The last inequality follows by squaring or from $W, C < \sqrt{D}$. Also $CD + 1 - A > 0$, so $G > A$, and $H > 1$ in either presentation. Consequently a_0, a_1 are positive on every allowed assignment.

The first strict majorant is immediate:

$$V_0^2 - a_0 = K^2 - 1 > 0. \tag{17}$$

Since $E = k_* DC$ and $(A^2 - 1)C^2 = D - 1$, we have $F = (D - 1)k_*^2 D^2 + 1$. The following exact identity supplies the second:

$$\begin{aligned} T_*^2 - 64DF &= (8D + 1)k_*^2 - 64D \\ &= 8D(k_*^2 - 8) + k_*^2 > 0. \end{aligned} \tag{18}$$

Indeed $k_* = iJC \geq 3$. Thus

$$\Delta_1 := V_1^2 - a_1 = \big[(8D + 1)k_*^2 - 64D\big](GH)^2 + 64DF(H^2 - 1) > 0. \tag{19}$$

The positive signs of V_0, V_1 matter: a squared inequality by itself would not justify the later positivity argument. The majorant (18) is inherited from the incoming sparse-Pell work.

5.3 A quotient shift makes the free positive C safe

The free coordinate may make Q negative. The offset W^2, already computed inside Q, repairs that problem with one extra addition:

$$J(n + W^2) + Q = Jn + (J - 2)W^2 + 3WC + 2 > 0. \tag{20}$$

For completeness we also need an upper bound on the translated quotient. Because $W \geq 1$ and $J \geq 3$,

$$Q/J \leq WC < D, \qquad W^2 + Q/J < 2D.$$

The elementary inequality $8D^2 - 4D - 1 \geq 3D$, together with $k_* \geq 3$, gives $T_* \geq 9D$. Hence $V_1 > 2D$ and

$$W^2 + Q/J < V_1. \tag{21}$$

No positivity of Q was assumed. In the original-H version, use $u = JF$ and $v = QF + J(H - C)$. Then $H - C > 0$, the translated numerator is still positive, and

$$W^2 + v/u = W^2 + Q/J + (H - C)/F < 2D + H < V_1.$$

5.4 Soundness: prove the margin first

Suppose $\mathcal{P} = 0$. If $m = 0$, then $\Omega = 0$ and $\mathcal{P} = [J(n + W^2) + Q]^4 > 0$, a contradiction. For $m \neq 0$, the factorization of the quartic norm over the reals provides signed real roots $r_0^2 = a_0$, $r_1^2 = a_1$ with

$$n + W^2 + Q/J = mZ_*, \qquad Z_* = V_0 + V_1 + r_0 + r_1. \tag{22}$$

The strict positive majorants imply $Z_* > 0$ even before either root is known to be rational. The left side is positive by (20); therefore $m > 0$.

Since $g > 0$ and $G_{\text{cap}} > 0$, the definition of m now yields

$$4(C - KY)^2 < K^2, \qquad C > K(Y - \tfrac{1}{2}) > 2K.$$

Because $K \geq P + R$ and $P > W$, this implies

$$C > 2(P + R) > 2R + 1 + W = B_1 + W. \tag{23}$$

Thus the missing old coordinate $\phi = C - B_1 - W$ is a positive integer. This is the structural saving: the polynomial did not have to build that lower bound into the expression for C.

To justify the radical merger, we must also exclude cancellation of two identical nonsquare radicals. We have

$$P < A^2, \qquad D > C^2, \qquad F \geq A^2, \qquad I \geq G^2 \geq A^2.$$

Together with $C > K$, these imply

$$DFI > A^4C^2 > P^2C^2 > P^2K^2 > a_0, \qquad a_1 = 64DFI > a_0. \tag{24}$$

Therefore $x = r_0 + r_1 = \Theta/\Omega$ is a nonzero rational number. The identity $2xr_0 = x^2 + a_0 - a_1$ makes r_0 rational, and then so is r_1. A rational number whose square is an integer is an integer: reduce its numerator and denominator to coprime integers. Hence both roots are integral, and $r_1/8$ is likewise an integer square root of DFI. Equation (22) now forces $J \mid Q$.

It remains to restore the old j parameter. Put $j_0 = j - 1 \geq 0$. The definitions give $F \equiv 1 \pmod{C}$ and $F \equiv 1 \pmod{J}$. Define

$$j_{\text{old}} = 1 + \frac{F - 1}{C} B_1 + F j_0 > 0.$$

This is an integer and gives $H = B_1 + j_{\text{old}}C$ and $F \mid H - C$. Since $\gcd(F, J) = 1$, the recovered $J \mid Q$ is exactly what is needed for the combined divisibility in (16). The source sizes, restored ϕ, square conditions, and positive margin now permit the Jones soundness theorem. It gives the carry conditions, which give a common source zero by Section 4.

5.5 Completeness: reconstruct the final quotient

Start from a common source zero. Choose a sufficiently large power-of-two b, encode g, and apply Jones completeness using $32g < G_{\text{cap}}$. Retain the old $C = B_1 + W + \phi$ as the new positive coordinate. The old combined divisibility, and $\gcd(F, J) = 1$, give $J \mid Q$ and $F \mid H_{\text{old}} - C$. Let $z = (H_{\text{old}} - C)/F$. Then $z > 0$, $z \equiv B_1 \pmod{C}$, and $0 < B_1 < C$; consequently $z = B_1 + j_0 C$ for a unique $j_0 \geq 0$. Take $j = j_0 + 1$. This gives exactly the new formula for H, without changing the radicands.

Choose positive integral roots r_0, r_1 of a_0, a_1, and set

$$n = m(V_0 + V_1 + r_0 + r_1) - W^2 - Q/J.$$

All quantities on the right are integers. By (21) and $m \geq 1$, the right side is strictly positive. The corresponding linear factor of the norm vanishes. This completes the proof of the source-system equivalence in Theorem 1.1.

Remark 5.1 (What the witness map asserts). Completeness chooses a new final quotient using positive square roots. It does not assert that subtracting W^2 from every old norm witness preserves positivity. Similarly, the CRT parameter is allowed to be zero internally; its positive-coordinate shift cannot be silently deleted.

Register	Degree	Register	Degree	Register	Degree
$D_{\mathrm{c}}(B)$	10380	R	22233	N	5932
A	45963	C	1	D	91928
F	367714	G	459643	H	389947
a_0	320246	a_1	2158822	m	184221
V_0	160123	V_1	1079411	Ω	230184
Δ_1	1883038	norm inner	2343406	$\mathcal{P}$	4686812

Table 2: Exact degrees certified by a full multivariate upper bound and an explicitly nonzero diagonal leading coefficient.

6 Exact degree and arithmetic size

6.1 Resolving the leading cancellation

The most important degrees of the primary, CRT-absorbed presentation are listed in Table 2. They refer to the nine unknowns only; the four source ports and the fixed scales have degree zero.

A naive maximum-and-sum degree propagation through (8) would overestimate its degree. Let $E_0 = J(n + W^2) + Q$. The exact identity

$$\begin{aligned}\Theta^2 - \Omega^2(a_0 + a_1) = E_0^2 - 2E_0\Omega(V_0 + V_1)\\ + \Omega^2\big[(K^2 - 1) + \Delta_1 + 2V_0V_1\big]\end{aligned} \tag{25}$$

uses (17) and (19). In the primary construction, $\Omega^2\Delta_1$ is the unique term of greatest degree on the right, namely 2 343 406. Its square has degree 4 686 812; the remaining term $4\Omega^4 a_0 a_1$ has strictly smaller degree. This gives a valid multivariate upper bound and isolates the leading form to be checked.

On the diagonal $b = g = h = s = w = C = i = j = n = t$, the source leaders are

$$[R]_{22233} = \lambda^{11115}(\lambda + 4), \qquad [N]_{5932} = \chi\lambda^{2966}.$$

The two terms S_{c} and T_{c} have the same degree and positive leaders; this is why $\lambda + 4$ appears after deleting the old leading block. Substituting into the positive defect in (19) and then (25) gives the exact diagonal leading coefficient

$$\boxed{2^{94}\lambda^{2\,343\,092}(\lambda + 4)^{104}\chi^{400} > 0.} \tag{26}$$

Thus the upper bound is attained for the actual fixed coefficients. This is an exact nonvanishing argument, not interpolation or sampling. The independent degree-certificate circuit uses the expanded defect identities, so unresolved top-degree cancellations cause verification to fail instead of being guessed away.

For a useful parametric check, let $r = \deg R$, $v = \deg N$, and let W_c be the maximum source weight. For either H presentation, the resulting degree in this range is

$$104r + 400v + 8W_c + 316.$$

Here $(r, v, W_c) = (22233, 5932, 183)$ gives the announced result. Keeping the original leading block would give 4 763 044; restoring the old composite C as well gives 6 692 108. These are checks on the contribution of each new step, not separate simplicity claims.

6.2 Shared evaluation of twenty quadratic rows

The degree improvements do not by themselves explain the 560-operation evaluator. For arithmetic size, reuse the common 60-operation quadratic source schedule to evaluate many of the rows in (9) together. Put $W_c = 183$, $u_0 = B^{W_c}$, and $u_i = B^{W_c - w_i}$. Evaluate each original positive source coordinate as $u_i + u_0$, aligning lower-degree additions by powers of u_0. For row r this computes the degree-two homogenization of p_r at $(u_0, u_1, \ldots, u_{31})$.

The factorial weight is two on a diagonal monomial, including the constant padding square, and one on a mixed monomial. Adding one further copy of each diagonal term therefore gives the required weighted row. For a target $t_r \geq 2W_c$, multiplication by $B^{t_r - 2W_c}$ produces that row's contribution to $D_{\mathrm{c}}(B)$ with no negative powers.

Exactly twenty targets meet this condition. The two low targets, 70 and 86, are evaluated directly from their sparse monomials. They cannot be forced through the homogeneous schedule using a negative exponent. This mixed evaluator preserves the unchanged selective code and gains sharing without moving any target. The independent checker expands all 22 source rows exactly, including the transport-coordinate adapter between the shared schedule and the authoritative original source.

The primary circuit has 332 multiplications, 173 additions, and 55 subtractions, for 560 operations. Every operation is live. Multiplication by a fixed numeral, every squaring, and every multiplication used to build a fixed power is charged. Reuse of an existing register is free. Fixed integer coefficients are input leaves, as in the repository's comparison metric; producing their binary payload is a separate finite preprocessing task. The certificate circuit is deliberately longer and is not used for the operation-count claim.

7 Proof status, reproducibility, and limits

7.1 What is proved here and what is imported

The free-coordinate and quotient-shift lemmas, the removal of the leading binary block, and the local short-equation synthesis are proved in this paper. The fixed selective code and sharpened Pell majorant are inherited from incoming submissions, with their identities checked again. The Jones ratio theorem is a separate numerical result with a repository Lean formalization. The interpretation of the fixed 31-variable system uses the repository's ordinary native-window compiler, coherent tuple adapter, and K215-to-syntactic-refutation calibration.

Those layers should not be conflated. The archive supplies no new end-to-end kernel proof of the 560-operation equation, and the finite audits cannot establish the universal compiler theorem. It also does not export a derivation inside the repository's object-language ZFC proof calculus. The stated ZFC-provable corollary is an ordinary mathematical composition of the identified arithmetic and compiler theorems. The active repository separately contains a closed, checked nine-variable construction; its proof status does not automatically certify this much smaller explicit circuit.

The distinction is relevant to incorporation: the new relative source-system equivalence can be formalized independently of the native compiler. Its hypotheses are just the explicit numerical source bounds and the existing Jones ratio interfaces. No new consistency assumption, oracle, nonconstructive choice of coefficients, or additional mathematical axiom is part of the construction.

7.2 Reproducible finite evidence

The package is self-contained for its numerical construction and audits; Python's standard library suffices. From the extracted archive root run

```
python3 verify_all.py
python3 -O verify_all.py
```

The driver checks the source payloads and declared fixed coefficient recipes, the full selective-isolation condition, the homogenized row identities, every operand and live gate, the Pell and norm identities, and the exact leading-degree certificate. The short-equation audit expands its complete leading homogeneous forms. Additional modular comparisons exercise separately compiled evaluators and are reported as transcription checks, not as universal proofs. Finite binary fixtures also check the two-block valuation identity, including its smallest boundaries.

The main files are the literal positive circuit, its independently arranged certificate circuit and degree ledger, the coefficient binding, the preserved source data and proofs, and the two short-equation circuits. The top-level README gives exact paths and evaluation commands. The hash manifest fixes the delivered bytes. The source pin is recorded so later repository changes cannot silently redefine the comparison.

To rebuild this paper from the archive root:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

7.3 Boundaries of the conclusion

These equations concern positive or explicitly translated nonnegative integer solutions. Substituting arbitrary integers for all inputs is not licensed by the proofs. A conversion to unrestricted integers would require a separate domain encoding and a new cost ledger.

The fixed coefficients remain very large, even though their compressed recipes and payloads are finite and reproducible. A 560-operation arithmetic circuit is not a claim of fast evaluation on large arbitrary inputs: bit complexity is a different measure. Finally, the note does not prove consistency, inconsistency, or an absolute minimum degree. Under the usual hypotheses of the second incompleteness theorem, a consistent ZFC cannot prove the displayed absence of zeros, since the equivalence would then prove its own consistency. No two-sided independence statement is inferred from consistency alone.

A Additional short equations: 73 operations at degree 54

The incoming archives contain useful improvements that are absent from the active P76 documentation: a 66-operation equation of degree 157, a 72-operation equation of degree 80, and an ordinary gap equation with 75 operations at degree 44. We retain these as the comparison baseline. The following synthesis adds the two tradeoffs

$$(\text{operations}, \text{degree}) = (73, 54), \qquad (74, 46),$$

each using fifteen strictly positive integer variables. It combines the incoming minimum-norm argument and squared-modulus reconstruction, and regroups the main norm to save one further operation. The finite verifier checks every literal arithmetic gate and exact leading homogeneous form. The semantic compiler remains an identified inherited mathematical dependency; no new proof-assistant or object-language ZFC certificate is asserted in this supplement.

A.1 The exact polynomial

Use the complete package-31 coefficient tuple, with $B = 2^{1064921}$, $\beta = B - 1$, and its already shifted M_F. The tuple and all four binary payloads are included in `short/coefficients31/`. The fifteen positive unknowns are

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, e, \gamma.$$

Define polynomial abbreviations, with no additional unknowns:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & & \\
X &= wq, & Y &= sq^3, & E &= XY, \\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\
H &= 4a + 3, & \Delta &= a^2 + H, & D &= X + ac + \gamma H, \\
L &= E(kY), & R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, & & \\
V &= f^2(cT - R) - c, & U &= \Delta(f^2 - 1), & I &= ic^2.
\end{aligned}$$

The five factors and strong residual are

$$\begin{aligned}
N_0 &= \tau^2 - L(L + k), & N_1 &= D^2 - \Delta c^2, \\
N_a &= (V + e)^2 - Ue(2V + e), & W_s &= I^2 - U, \\
N_k &= k - hE - R, & N_t &= (K_0 + w)C + (q - F) - z(q - 1).
\end{aligned}$$

The main short equation is

$$\boxed{Q_{54} = N_0^2 + N_a^2 + W_s^2 + (N_1 N_k N_t - 1)^2 - 2 = 0.} \tag{27}$$

The lower-degree alternative is

$$\boxed{Q_{46} = N_0^2 + N_1^2 + N_a^2 + W_s^2 + (N_k N_t - 1)^2 - 3 = 0.} \tag{28}$$

The subscripts denote exact degree. Their literal evaluators have 73 and 74 operations, respectively. The principal evaluator has 39 multiplications, 19 additions and 15 subtractions; the alternative has one more addition. Multiplication by a fixed numeral is charged.

A.2 The minimum argument and complete local equivalence

The following facts hold on every positive assignment, before imposing either equation:

$$N_0^2 \geq 1, \qquad N_a^2 + W_s^2 \geq 1. \tag{29}$$

Equality cases are $N_0 = 1$ and $(N_a, W_s) = (1, 0)$, respectively. For the first assertion, write $n = XY^2 \geq 2$. The discriminant $n(n+1)$ is nonsquare, so $N_0 \neq 0$. A hypothetical solution of $x^2 - n(n+1)t^2 = -1$ has $nt < |x| < (n + \frac{1}{2})t$. Multiplication by $2n + 1 - 2\sqrt{n(n+1)}$ produces a new positive second coordinate $(2n+1)t - 2|x| < t$, contradicting minimality. Thus $N_0 \neq -1$.

For the second assertion put $y_0 = V + e$. The exact identity

$$N_a = UV^2 - (U - 1)y_0^2$$

is immediate. If $U = 0$, then $W_s = I^2 \geq 81$. Otherwise $U \geq 2$, and $N_a = 0$ would force the consecutive integers $U, U - 1$ to be squares, unless $V = y_0 = 0$. The latter contradicts $e > 0$. Hence

$N_a \neq 0$. Equality in the second block forces $W_s = 0$ and $N_a = \pm 1$. Now $U = I^2$, so modulo four N_a is congruent to y_0^2 when I is even and to V^2 when I is odd. Thus $N_a = -1$ is impossible. This proves all assertions in (29).

At a zero of (27), both minimum blocks attain their minima and the remaining square vanishes. Hence

$$N_0 = N_a = 1, \qquad W_s = 0, \qquad N_1 N_k N_t = 1.$$

The last equation makes N_1 an integer unit. Since $\Delta = (a+2)^2 - 1 \equiv 0$ or $3 \pmod 4$, square residues exclude $N_1 = -1$. Thus $N_1 = 1$ and $N_k N_t = 1$. These are precisely the equality conditions of (28); conversely they imply both equations. This is a complete tuplewise equivalence, uniform in the fixed ports with $\beta > 0$.

The saving is transparent at the circuit level. Starting from Q_{46}, remove the N_1^2 gate and its summation gate, insert one multiplication of N_1 into the unit product, and change the final constant from three to two. This saves exactly one binary operation.

A.3 Squared-modulus completeness for the ordinary strong form

The historical gap construction used $V_{\text{old}} = c(T_{\text{old}} f - 1) - Rf^2$. The identity

$$V = f^2(cT - R) - c = V_{\text{old}}\big|_{T_{\text{old}} = fT}$$

proves soundness immediately. Completeness requires reconstructing an auxiliary Pell ordinate, rather than assuming divisibility of every old quotient.

A halting computation supplies a canonical normalized witness. Transfer it to ordinary coordinates by $i_{\text{ordinary}} = \Delta i_{\text{normalized}}$. The inherited decoding supplies $c = \psi_A(R)$, $A = a + 2$, $R \equiv 3 \pmod 4$, and

$$I^2 = \Delta(f^2 - 1), \qquad c \mid I, \qquad f^2 \equiv 1 \pmod c, \qquad \gcd(c, f) = 1.$$

For $R = 2r + 1$ the Pell polynomials satisfy

$$\chi_t(R) = tQ_r(t^2), \quad Q_r(0) = (-1)^r R, \quad Q_r(1 - A^2) = (-1)^r \psi_A(R).$$

These follow by the elementary second-order Pell recurrences. Since r is odd, take

$$V = \frac{\chi_I(R)}{I}, \qquad y = \psi_I(R).$$

Both are positive integers, $V > 1$, and their norm is one. The displayed polynomial identities give $V \equiv -c \pmod{f^2}$ and $V \equiv -R \pmod c$. Consequently

$$T = \frac{V + c + Rf^2}{cf^2} > 0$$

is an integer and satisfies the required quotient formula. Finally

$$(I^2 - 1)(y^2 - V^2) = V^2 - 1 > 0$$

implies $e = y - V > 0$. The other coordinates remain unchanged. This proves existence equivalence to the authenticated ordinary source and hence, using its inherited compiler and K215 calibration, the ordinary mathematical implications formalizable in ZFC:

$$Q_{54} \text{ has no positive zero} \longleftrightarrow \text{Con(ZFC)},$$
$$Q_{46} \text{ has no positive zero} \longleftrightarrow \text{Con(ZFC)}.$$

A.4 Exact degrees and bounded comparison

The factor degrees are $22, 18, 23, 22, 7, 2$ for $N_0, N_1, N_a, W_s, N_k, N_t$, respectively. The main-norm cancellation is the identity, with $Z = X + \gamma H$,

$$(ac + Z)^2 - (a^2 + H)c^2 = Z^2 + 2acZ - Hc^2.$$

Writing $Q = \beta J$, $k_0 = \eta + \zeta$ and $C_1 = Q - F - \alpha$, the highest homogeneous forms are

$$[Q_{54}]_{54} = 64\gamma^2 h^2 w^6 s^8 k_0^2 Q^{30}(wC_1 - zQ)^2,$$
$$[Q_{46}]_{46} = 4w^4 s^6 k_0^2 T^2 f^8 e^2 Q^{22}.$$

Their respective coefficients of $J^{32}\gamma^2 h^2 w^6 s^8 \eta^2 z^2$ and $J^{22} w^4 s^6 \eta^2 T^2 f^8 e^2$ are $64\beta^{32}$ and $4\beta^{22}$, strictly positive for the supplied tuple. The exact symbolic audit verifies these full leading forms, not merely a numerical specialization.

These points improve the local degree/operation frontier between the reviewed 72/80 and 75/44 candidates. They do not improve the existing 66-operation record, the degree-44 point, or the smallest-variable constructions. The proof files and source receipts make this a bounded repository comparison, without a claim of worldwide priority or global minimality.

B The authoritative 31-coordinate source

This appendix fixes the source completely and avoids a possible transport-coordinate ambiguity. Here the names are local source coordinates, separate from the nine inputs in the main construction. Their order is

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta).$$

All are positive in the source. Define the affine abbreviations $q = \beta J + 1$, $k = \eta + \zeta$, $H_s = 4a + 3$, and $C_s = q - F - \alpha$. The 22 equations, in their authoritative order, are

$$L = q^2 - 1, \qquad N = q(L + 1),$$
$$X = wq, \qquad a = E + sN,$$
$$E = X(a - E), \qquad c = k(a - E) + \eta,$$
$$P = E(c - \eta) + k, \qquad \tau^2 = P(P - k) + 1,$$
$$D = X + ac + \gamma H_s, \qquad \Delta = a^2 + H_s,$$
$$C_2 = c^2, \qquad D^2 = \Delta C_2 + 1.$$

The next three rows are

$$k = jc - V + 1 + hE,$$
$$jc - V = (q - F)(N - q) - C_s L + (M_C + qM_F)J,$$
$$(K_0 + w)C_s + (q - F) - z(q - 1) = 1.$$

The final seven rows are

$$F_2 = f^2, \qquad V = of - c,$$
$$g = y^2 - V^2, \qquad I = iC_2,$$
$$B_{\mathrm{aux}} = I^2, \qquad B_{\mathrm{aux}} = \Delta(F_2 - 1),$$
$$y^2 - B_{\mathrm{aux}} g = 1.$$

Take left minus right, and replace each of these 31 coordinates by $x_i + 1$, to obtain $p_1, \ldots, p_{22}$. Each residual has degree at most two in $\mathbf{x}$ after the fixed ports are specialized. In particular, the original positive transport coordinate is used in row 15. The shifted transport equation $(K_0+w)C_s = F+z\beta J$ belongs to a related source but is not substituted into this definition.

The archive retains the original JSON and its historical metadata unchanged. The active manifest explicitly selects the whole package-31 tuple; old metadata in preserved evidence files does not override it. The shared evaluator's transport adjustment is checked symbolically against this exact source.

C Provenance and an incorporation checklist

This note was derived from the placed collection and incoming submissions at the pin given in Section 1. The principal immediate sources are:

- `ZFC_Diophantine_Selective_Quadratic_2026-10-23.zip`: the exact original 31-variable source, selective weights and targets, sparse target padding, and parity-calibrated threshold.
- `con-zfc-sparse-pell-124c07c0b7.zip`: the positivity-first radical merger, CRT absorption, and the majorant $k_*(8D^2 - 4D - 1)$.
- `ConZFC_Sparse_Support_2026-10-23.zip`: the ordinary gap minimum argument and its coherent coefficient binding.
- `con_zfc_sparse_arithmetic_2026-10-23.zip`: the squared-modulus Pell reconstruction used in Appendix A.

The source receipts, copied notices, and licenses are retained. Newly authored material is marked separately; no claim is made to relicense inherited files. The independent reviews were performed within this AI-assisted session and are not external human endorsements.

For incorporation, the mathematical milestones can be separated cleanly: first prove the abstract shifted-norm/free-coordinate lemma; then the two-block valuation formula and threshold identity; then certify the literal shared evaluator against the sparse source; finally connect that construction to the existing compiler endpoint. Each step has a precise statement and finite data boundary in the archive. The present note does not modify the repository or claim that those future kernel proofs have already been compiled.

References

[1] James P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic **47** (1982), no. 3, 549–571. https://doi.org/10.2307/2273588.

[2] Vladimir Reshetnikov et al., *ProveIt*, snapshot `4b4349064a750ba6347293b55874a7f92abdd0db`, accessed 23 October 2026. Pinned repository source.

[3] ProveIt incoming research archives, same snapshot. Pinned incoming collection.

[4] ProveIt, *NineRatio.lean*, numerical Jones ratio interfaces, same snapshot. Pinned numerical theorem source.

[5] ProveIt, *A closed executable equation for ZFC consistency*, `EXPLICIT_EQUATION.md`, same snapshot. The separate checked construction and its object-language formalization boundary. Pinned endpoint and proof-status document.